

Comparison criterion	Scenario A: Full replacement	Scenario B: Component repair and replacement	Initial interpretation	Main related code
Whole-window material demand	Higher, because full replacement requires a complete new window unit	Lower, if the long-life frame is retained and only shorter-life parts are replaced	Scenario B is likely preferable if the retained frame remains functional and accessible	Higher initial material use
Operational performance improvement	Potentially higher if the new replacement window significantly improves thermal or acoustic performance	Depends on whether repaired components maintain required performance or whether glazing can be upgraded	Scenario A may be stronger if the existing system performs poorly, but this must be checked	Lifecycle verification
Repair and maintenance burden	Lower routine repair burden before replacement, but replacement is a major intervention	Higher number of smaller interventions over time	Scenario B may avoid full replacement but requires practical maintenance planning	Maintenance planning
Transport and service visits	Transport for full replacement, removal and installation	Transport for multiple repair or service visits	Benefit depends on distance, frequency and whether repairs are grouped efficiently	Additional transport
Spare-part dependence	Lower after full replacement because new system resets compatibility	Higher, because repair depends on compatible parts over time	Scenario B depends on supplier commitment and long-term parts availability	Spare parts
Component access	Less important before replacement, but important for removal and end of life	Critical, because inaccessible components can make repair impractical	Poor access can turn Scenario B into an implementation gap	Component accessibility
End-of-life burden	Higher at replacement point because old full window must be removed and treated	Delayed full-window end of life, but smaller components still create waste over time	Scenario B may reduce premature full-window disposal but still needs end-of-life planning	Burden shifting
Reuse potential	Limited because removed windows are made to measure	Limited, but retained frame avoids needing direct reuse of full unit	Both scenarios face reuse barriers due to dimensions and certification	Direct reuse
Risk of premature replacement	Higher if replacement is chosen because the product feels newer or more modern rather than because the old system failed	Lower if repair-first logic is followed, but still possible if parts are unavailable or access is poor	Interviews suggest premature replacement can be driven by client expectations and practical repair barriers	Premature replacement
Technological lock-in	Possible if replacement creates a high-performance but difficult-to-upgrade system	Possible if long-life frame restricts future glazing or performance upgrades	Both scenarios need future upgrade compatibility checks	Lock-in
Overall lifecycle confidence	Medium to low without quantified embodied, operational and end-of-life data	Medium to low without repair-frequency, spare-part and service-life data	Neither scenario should be claimed as automatically better without assumptions and uncertainty ranges	Lifecycle verification